MIL-L-81787B(AS)
9 June 1983
SUPERSEDING
MIL-L-81787A(AS)
1 April 1975

MILITARY SPECIFICATION

LIFE PRESERVER, INFLATABLE, UTILITY, TYPE LPP-1A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for one type of inflatable utility life preserver designated as Type LPP-1A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-268	-	Paper, Kraft, Wrapping.
UU-T-81	-	Tags, Shipping and Stock.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-76	-	Tape, Packaging, Paper, (For Carton Sealing).

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U. S. Military Property.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

68A94C1	-	Part Index, Utility Inflatable Life Preserver, Type LPP-1A.

(Copies of specifications, standards and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The Type LPP-1A utility inflatable life preservers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, 6.3 and 6.3.1).

3.2 First article. When specified (see 6.2), a sample of the Type LPP-1A life preserver shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Materials and components. The materials and components shall conform to applicable specifications, standards and drawings as listed or required herein. Unless otherwise specified, the manufacturer shall certify that the materials and components, except for the metallic parts used in the construction of the life preservers shall have been manufactured not more than 18 months prior to the date of delivery of the life preservers (see 6.2.2).

3.4 Design and construction. The design and construction of the LPP-1A life preserver shall be in accordance with drawings listed on Drawing 68A94C1.

3.4.1 Cut edges. All the cut edges of the uncoated nylon tapes (including the nylon fastener tapes), webbings, and cords shall be seared, prior to the fabrication of the life preserver, to prevent fraying. No sharp edges shall be formed.

3.4.2 Use of the adhesive. The adhesive used shall be in accordance with applicable drawing requirements. The color of the adhesive shall be natural (un-

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pigmented). In all the cementing operations, the surface to which the adhesive is to be applied shall be thoroughly clearned with a suitable solvent in such a manner that the dusting materials (zinc stearate or talc) or any other surface contaminant are removed. Care should be exercised to ensure that the coating and the base undercloth are not damaged and the adhesion between both is not impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive and shall leave no residue. The adhesive shall be controlled to ensure that old adhesive or adhesive which has partly or completely polymerized is not used. A fresh batch of the adhesive shall be used at least every eight hours during the course of manufacture (see 4.5.1.1.3). The containers for the adhesive shall be free from congealed adhesive before being refilled.

3.4.3 Cementing of the seams, seam tapes, patches and attachments. Seam construction and seam tape, patch, and attachment applications shall be in accordance with applicable drawing requirements. The techniques and precautions outlined in 3.4.2 shall be used in order to effect optimum bonding properties. The adhesive shall be applied in a straight line parallel to the edges of the seam, seam tape, patch or attachment; and shall extend from just being visible to a maximum of 3/8 inch beyond the applicable edge. The ends of seam tapes shall overlap 3/4, plus 1/2, minus 0, inch. Cementing shall be performed in a manner to prevent clotting, localized stiffness, shrinking, puckering, spotting or staining. The cemented areas shall not contain any voids, open channels, edge lifts, or excessive wrinkling.

3.4.4 Seams and stitching. No stitching shall pass through any air retaining compartment. Sewing shall only be used in the construction of the accessory parts but not in their final attachment to the life preserver. All the machine stitching in the accessory parts shall be accomplished as specified in the applicable drawing. All the sewing shall be accomplished with stitch Type 301 conforming to FED-STD-751, eight to ten stitches per inch. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching in any row shall be maintained within a tolerance of plus or minus 1/32 inch. The ends of the stitching shall be backstitched, by overlapping on itself, a minimum of 1/2 inch. Thread breaks, skips and run-offs shall be overstitched not less than one inch. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together. No seam shall be twisted, puckered or pleated and no portion of the accessory parts shall be caught in an unrelated operation or seam. All the thread ends shall be trimmed to a length of 1/4 to 3/8 inch. The seam edges shall be properly forced out and shall not contain any folds.

3.4.4.1 Bartacking. The number of stitches per bartack shall be based proportionally on 1/4 inch long bartack containing a minimum of 14 stitches. The length and location of each bartack shall be as specified in the applicable drawing.

3.5 Color. The color of the LPP-1A preserver and the components shall be as specified on the applicable drawings.

3.6 Markings. Markings shall be in accordance with MIL-STD-130 and applicable drawing requirements. Ink, color, size and location of markings shall be as specified on applicable drawings. Stenciling and/or silk screening methods of application are acceptable, except that markings 1/4 inch or less in height shall not be stenciled. All markings shall be thoroughly dry prior to packaging.

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3.6.1 Warning tag. A red warning tag conforming to UU-T-81, type B, not smaller than size 4, and strung with twine, shall be securely tied to the buckle of the belt assembly. The tag shall protrude from the storage container and shall contain the following markings in black letters and numerals:

WARNING

THIS LIFE PRESERVER SHALL NOT BE USED
UNTIL SUBJECTED TO THE CALENDAR
INSPECTION OF NAVAIR 13-1-6.1, AVIATION-
CREW SYSTEMS MANUAL.

3.6.2 Serial numbers. The preservers shall be identified by individual serial numbers which shall be assigned by the manufacturer (see 6.2.1i). Serialization shall be by a block of consecutive numbers to cover the entire acquisition document quantity. Method of marking shall be as specified in 3.6 and applicable drawings.

3.7 Performance characteristics.

3.7.1 Weight. When inspected as specified in 4.8.1, the weight of the life preserver assembly shall not exceed 3.00 pounds.

3.7.2 Operation (carbon dioxide). When the life preserver is inflated for operations test as specified in 4.8.3, design shape shall be attained within thirty seconds. There shall be no leakage of carbon dioxide, no hindrance to the flow of carbon dioxide or evidence of material or constructional failure in any respect. The seams of attachments shall not pucker when the compartment is inflated or deflated. Upon completion of this test and examination, and while still inflated with carbon dioxide, the preserver shall be subjected to the buoyancy test (3.7.3 and 4.8.4) if required.

3.7.2.1 Oral inflation assembly operation. When inspected as specified in 4.8.3 (or 4.8.4 when applicable), there shall be no hindrance to the flow of carbon dioxide when the valve is depressed and a positive shut-off of the flow of carbon dioxide when the valve is released. Additionally, there shall be no evidence of material or constructional failure in any respect.

3.7.3 Buoyancy. When inspected as specified in 4.8.4, the life preserver, while inflated with the carbon dioxide, shall support a 29 pound steel weight without the entire life preserver assembly sinking below the surface of the water.

3.7.4 Pressure. When inspected as specified in 4.8.5, the pressure in the flotation compartment shall be not less than 4.5 pounds per square inch gage (psig). There shall be no evidence of constructional or material failure in any respect. The seams or attachments shall not pucker when the compartment is inflated or deflated.

3.7.5 Leakage. When inspected as specified in 4.8.6, the pressure in each compartment shall be not less than 1.60 psig. There shall be no evidence of constructional or material failure in any respect. The seams or attachments shall not pucker when the compartment is inflated or deflated.

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3.7.6 Visual and dimensional examinations.

3.7.6.1 Visual. When examined as specified in 4.8.2.1, the preservers shall conform to the requirements of this specification and applicable drawings. The classification of defects for visual examination of the life preserver (Table IV) shall be used to classify the defects found.

3.7.6.2 Dimensional. When examined as specified in 4.8.2.2, the preservers shall conform to the requirements of this specification and applicable drawings. The list of defects for dimensional examination of the life preserver (Table V) shall be used to enumerate the defects found.

3.8 Insertion of the life preserver into the pouch and storage container. Upon satisfactory completion of all tests and examinations, the life preserver shall be dusted with a minimum amount of zinc stearate or talc to prevent the adjacent cloth surfaces from adhering. Prior to the insertion of the life preserver into the pouch, the flotation compartment shall be completely evacuated of air and the oral inflation valve shall be locked in the closed position by the knurled ring. The flotation tube sides shall then be folded over to approximate the width of the pouch and rolled into the pouch, after which the pouch shall be snapped closed. The pouch, containing the life preserver, shall be inserted into the storage container and the storage container shall be snapped closed.

3.9 Workmanship. After completion of the final assembly, the life preservers shall be thoroughly cleaned and all loose thread, lint, and foreign matter shall be removed. The life preservers shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability or durability. The life preservers shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Quality conformance verification inspection (see 4.6).

4.3 Qualification.

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4.3.1 Samples. The qualification inspection samples shall consist of two life preservers. The qualification samples shall be forwarded as specified in 6.3. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples for qualification inspection
LIFE PRESERVER, INFLATABLE, UTILITY, TYPE LPP-1A
Manufacturer's designation or number
Name of manufacturer
Submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-L-81787B(AS) under authorization (reference authorizing letter and number) (see 6.3)

4.3.2 Inspection. The qualification inspection of the LPP-1A life preservers shall consist of the examinations and tests specified in Table I. Listed sequence is mandatory.

4.3.3 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. The time of periodic verification by certification shall be in two-year intervals from the date of the original qualification. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.4 First article.

4.4.1 First article samples. Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacturer shall submit three life preservers. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these life preservers from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1e). Approval of the first article samples or the waiving of the first article inspection does not exempt the requirements for performing the quality conformance inspection. The first article samples shall be forwarded as specified in the contract or order (see 6.2.1f).

4.4.2 First article inspection. The first article inspection of the Type LPP-1A life preserver shall consist of the examinations and tests specified in Table II. The examinations and tests shall be performed in the sequence listed.

4.4.3 First article approval. Upon completion of the first article inspection program, pertinent comments and recommendations will be forwarded by the Government activity responsible for conducting the inspection program (see 6.2.1f) to the contracting officer. One approved life preserver will be returned to the manufacturer for use in monitoring production. One life preserver will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract. The other life preserver will be retained by the inspection laboratory for reference and will be returned to the manufacturer with the samples from the final lot as required by 4.6.3.

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4.5 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Weight of the life preservers
Visual examination of the life preservers
Operation (carbon dioxide)
Buoyancy
Pressure
Leakage
Dimensional check of the life preservers
Packaging

4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1.1.1 Life preservers. An inspection lot size shall be expressed in units of one life preserver made essentially under the same conditions and from the same materials and components. The sample unit shall be one life preserver.

4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing life preservers, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing life preservers, fully prepared for delivery with the exception that it need not be sealed.

4.5.1.1.3 Adhesive. An inspection lot size shall be expressed in units of one batch of the adhesive. An inspection lot shall consist of all the batches of the adhesive used by the manufacturer during one day's production of the preservers (see 3.4.2). The sample unit shall be one batch of the adhesive used during one day's production of the preservers.

4.5.1.2 Sampling for tests and examinations of the life preservers and packaging. The sample size, acceptance criteria, tests, and examinations required for the life preservers or the packaging shall be as specified in Table III.

4.6 Quality conformance verification.

4.6.1 Quality conformance verification samples. Upon completion of the tests and examinations specified in 4.5.1.2, a random sample shall be selected from each lot in accordance with MIL-STD-105, Inspection Level S-3 (see 6.2.1h). The sample size shall be based only upon the applicable sample size code letter corresponding to Inspection Level S-3. Each preserver selected as a sample unit shall be identified by its assigned serial number (see 3.6.2). Additionally, the serial numbers of the units within the lot, represented by the sample units shall be furnished to the inspection facility. The samples shall be forwarded to the inspection facility specified in the acquisition document (see 6.2.1g).

4.6.2 Quality conformance verification inspection. The quality conformance verification inspection shall consist of the following tests and examinations, listed sequence mandatory:

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Tests and examinations

Weight	(4.8.1)
Operation	(4.8.3)
Pressure	(4.8.5)
Leakage	(4.8.6)
Visual examination	(4.8.2.1)
Dimensional examination	(4.8.2.2)

4.6.3 Quality conformance verification approval. Upon completion of the quality conformance verification inspection (see 4.6.2), the Government activity responsible for conducting the inspection program (see 6.2.1g) shall report the result of tests, with recommendations, to the contract administration office specified in the acquisition document. Final acceptance of the lot from which the samples were selected shall be based upon successful completion of the inspection program by the cognizant Government quality assurance representative specialist; applying the applicable acceptance criteria specified in Table III. Upon completion of this inspection program the samples will be returned to the contractor.

4.7 Inspection conditions.

4.7.1 Atmospheric conditions. Unless otherwise specified, all the inspections required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches (711.2 to 812.8 mm) of mercury and at a temperature of 77 ± 18 degrees Fahrenheit (25 ± 10 degrees Celsius). If the final values of the ambient temperature or barometric pressure at the end of the four hour air pressure inspections are different from the initial values recorded at the start of the inspection, the following corrections shall be made to the final pressure readings in psig.

4.7.1.1 Temperature correction. For each degree Fahrenheit rise in temperature, 0.031 psig shall be subtracted from the final pressure reading. For each degree Fahrenheit drop in temperature, 0.031 psig shall be added to the final pressure reading. The corresponding correction per degree Celsius is 0.056 psig.

4.7.1.2 Barometric pressure correction. For each 0.1 inch of mercury rise in barometric pressure, 0.049 psig shall be added to the final temperature-corrected pressure reading. For each 0.1 inch of mercury drop in barometric pressure, 0.049 psig shall be subtracted from the final temperature-corrected pressure reading.

4.7.2 Pressure measurement. The pressure shall be measured by means of a mercury manometer or low pressure gauge calibrated in tenths psig or fifths of inches of mercury. Inches of mercury can be converted to psig by multiplying the inches of mercury by 0.49.

4.7.3 Inspection area and equipment. The area, in which the life preservers are inspected, shall be adequately protected to preclude damage to the units. The area and inspection equipment shall be free of sharp or rough edges, burrs, protrusions, etc., which will cut, tear or damage the life preservers or components.

4.7.4 Air. When use of air is specified in an inspection, the air used shall not contain any oil or condensed water vapor.

4.8 Inspection methods.

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4.8.1 Weight. The weight ofthe life preserver (belt assembly and pouch attached) with storage container and the carbon dioxide filled cylinders shall be determined on a scale capable of weighing to the nearest 0.01 pound. The weight shall conform to the requirements specified in 3.7.1.

4.8.2 Visual, dimensional and packaging examination.

4.8.2.1 Visual. Every preserver shall be visually examined for defects to determine conformance to the requirements of 3.7.6.1.

4.8.2.2 Dimensional. Every preserver shall be dimensionally examined to determine conformance to the requirements of 3.7.6.2.

4.8.2.3 Packaging. Each of the fully prepared shipping containers, containing life preservers, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table VI, shall be used to enumerate the defects found.

4.8.3 Operation (carbon dioxide). The preserver, removed from its storage container and pouch, shall be spread out and placed in a flat condition on a table in the inspection area (see 4.7.3). The preserver, with the specified carbon dioxide filled cylinder installed, shall be inflated by pulling the lanyard with a quick downward movement and observed for conformance to the requirements of 3.7.2.

The use of inflation assemblies solely for inspection purposes is prohibited. Test carbon dioxide cylinders (refilled cylinders) may be used for this test provided they conform to applicable drawing and specification requirements. If used, test cylinders shall be clearly and permanently marked to be readily identifiable as such.

Unless otherwise specified, immediately upon completion of this test and examination the buoyancy inspection (see 4.8.4) shall be performed while the preserver is inflated with carbon dioxide.

If the buoyancy inspection is not required (see 4.6.2), the preserver shall be completely deflated by unlocking the oral inflation valve, depressing the mouthpiece and discharging the carbon dioxide. During the discharge of carbon dioxide, the mouthpiece shall be released a minimum of three times and observed for conformance to the requirements of 3.7.2.1.

4.8.4 Buoyancy. The buoyancy inspection, when required, shall be performed in conjunction with the operation inspection (see 4.8.3) while the preserver is still inflated with carbon dioxide. The preserver, with a 29 pound steel weight attached, shall be placed in clean, fresh water having a temperature of 75 ± 5° Fahrenheit (23 ± 3° Celsius), and observed for conformance to the requirements of 3.7.3.

Upon completion of the test and inspection, the preserver shall be completely deflated through its oral inflation assembly in the manner outlined in 4.8.3 and observed for conformance to the requirements of 3.7.2.1.

The preserver shall then be dried without the application of heat in preparation for the pressure inspection (4.8.5).

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4.8.5 Pressure. The carbon dioxide cylinder shall be removed for this inspection. The preserver shall be inflated with air (see 4.7.4) through its oral inflation assembly to a pressure of 5 psig. The air shall then be securely shut off and after a minimum of 15 minutes the pressure shall be checked and readjusted, if necessary, to the original pressure of 5 psig. Fifteen minutes after the check and readjustment, the preserver shall be observed for conformance to the requirements of 3.7.4.

Upon completion of this test and inspection, the preserver shall be completely deflated in preparation for the leakage inspection (4.8.6).

4.8.6 Leakage The carbon dioxide cylinder shall be inflated with air (see 4.7.4) through its oral inflation assembly to a pressure of 2 psig. The air shall then be securely shut off and after a minimum of 15 minutes the pressure shall be checked and readjusted, if necessary, to the original pressure of 2 psig. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 4 hours after the check and readjustment, the pressure shall be measured and corrected for any change in the temperature and barometric pressure (see 4.7 through 4.7.1.2). The preserver shall conform to the requirements of 3.7.5. Upon completion of the inspection, the life preserver shall be completely deflated.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or Commercial, as specified (see 6.2.1).

5.1.1 Level A. Each life preserver, in its pouch and storage container, prepared as specified in 3.8, shall be completely wrapped in 30 pound minimum basis kraft paper conforming to UU-P-268. The wrapped preserver shall be packaged within a snug fitting fiberboard container conforming to PPP-B-636, Style CSSC, Type CF or SF, Weather-Resistant Class, Variety SW, Grade W5c or W5s. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches. All the seams and joints shall be sealed with water resistant tape conforming to PPP-T-76, 2 inches wide minimum.

5.1.2 Commercial. Each life preserver in its pouch and storage container, prepared as specified in 3.8, shall be individually packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2.1). Shipping containers, insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight.

5.2.1 Level A. Six life preservers, packaged as specified in 5.1.1 or 5.1.2, shall be packed as specified in 5.2.2, except that the fiberboard container shall be Weather-Resistant Class, Variety SW, Grade V3c or V3s. In addition, each container shall be reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Six life preservers, packaged as specified in 5.1.1 or 5.1.2, shall be packed edgewise into a close fitting fiberboard container conforming to

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PPP-B-636, Style CSSC, Variety SW, Grade 275, type-optional. The container body joint and top and bottom flaps, closure thereof shall be accomplished by glueing. The container shall be reinforced (strapped) by two non-metallic bands, applied in accordance with applicable procedures in the appendix to PPP-B-636. No metal staples or stitches are permitted on the container.

5.2.3 Commercial. The packaged life preserver shall be packed within exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. The container shall be reinforced (strapped) with two non-metallic bands. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special markings required by the contract or order, the interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

6. NOTES

6.1 Intended use. The LPP-1A life preserver is intended for use by passengers in cargo or transport type aircraft for sea survival situations and by helicopter aircrewmen, as required, with body armor.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification, including any amendments.

b. Applicable Qualified Products List (see 4.3 and 6.3).

c. Government part number, national stock number, and quantity.

d. Applicable drawings listed on Index Drawing 68A94C1, including revisions.

e. First article inspection may be waived (see 4.4.1).

f. Name and address of the first article inspection laboratory (see 4.4.1); and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.4.3).

g. Selection of quality conformance verification samples to be performed by the cognizant Government quality assurance specialist (see 4.6.3).

h. Name and address of the quality conformance verification inspection laboratory (see 4.6.1); and the name and address of the Government activity responsible for conducting the quality conformance verification inspection program (see 4.6.3).

i. Serial numbers to be assigned by the manufacturer (see 3.6.2).

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j. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

k. Whether any special markings are required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement	Applicable DID no.
3.3	Certificate of compliance for the age of materials and components.	DI-E-2121
4.4.3	First article inspection reports	DI-T-5329

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Qualification. With respect to products requiring qualification awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL 81787) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Warminster, PA 18974, Attention: Code 6031. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a date on which the samples can be forwarded and requesting an authorization number to accompany the samples plus the name and address of the qualification inspection facility.

6.3.1 Qualification inspection report. When requested, the manufacturer shall submit an inspection report in accordance with SD6, Provisions Governing Qualification.

6.4 First article. When a first article is required, the Type LPP-1A preserver will be tested and should be a first production item. The first article should consist of three units. The contracting officer should include specific instructions

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in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - AS

(Project No. 4220-N282)

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TABLE I. Qualification examinations and tests (see 4.3.2).

Inspection	Paragraph	
	Requirement	Method
Weight of life preserver	3.7.1	4.8.1
Operation (carbon dioxide)	3.7.2	4.8.3
Buoyancy	3.7.3	4.8.4
Pressure	3.7.4	4.8.5
Leakage	3.7.5	4.8.6
Visual examination	3.7.6.1	4.8.2.1
Dimensional check	3.7.6.2	4.8.2.2

TABLE II. First article examination and tests (see 4.4.2).

Inspection	Paragraph	
	Rquirement	Method
Weight of life preserver	3.7.1	4.8.1
Operation (carbon dioxide)	3.7.2	4.8.3
Buoyancy	3.7.3	4.8.4
Pressure	3.7.4	4.8.5
Leakage	3.7.5	4.8.6
Visual examination	3.7.6.1	4.8.2.1
Dimensional check	3.7.6.2	4.8.2.2

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TABLE III. Sample size, acceptance criteria, tests, and examinations of the life preserver and packaging (see 4.5.1.2).

Inspection 1/	Paragraph		Sample Size 2/	Acceptance criteria
	Requirement	Method		
Weight	3.7.1	4.8.1	Inspection Level S-3	An acceptable quality level of 1.5 defects per 100 units
Visual examination	3.7.6.1	4.8.2.1	Every life preserver for critical defects. Inspection Level II for minor defects	Reject all units with any critical defect, and an acceptable quality level 15.0 defects per 100 units for minor defects
Operation (carbon dioxide) 3/	3.7.2	4.8.3	Every life preserver	Reject all defective units
Buoyancy 3/	3.7.3	4.8.4	Inspection Level S-3	An acceptable quality level of 1.0 defects per 100 units
Pressure 3/ 4/	3.7.4	4.8.5	Every life preserver	Reject all defective units
Leakage 3/	3.7.5	4.8.6	Every life preserver	Reject all defective units
Dimensional check	3.7.6.2	4.8.2.2	Inspection Level S-3	An acceptable quality level of 10.0 defects per 100 units
Packaging	Section 5	4.8.2.3	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units

1/ The results of inspection shall be identifiable by the assigned serial numbers (see 3.6.2).

2/ The sample size shall be based upon the applicable sample size code letter corresponding to the specified level of MIL-STD-105.

3/ These inspections shall be performed in listed sequence.

4/ Pressure inspection may be performed prior to final assembly of the life preserver. If not performed prior to final assembly, footnote 3/ shall be mandatory.

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TABLE IV. Classification of defects for visual examination of the life preserver (see 4.8.2.1).

Critical	Minor
1. Any non-specified hole, cut, patch, or burn.	201. Stitching and sewed seam construction, not as specified. 3/
2. Any fabric damaged, bruised, abraded, containing imperfections, or is otherwise defective. 1/	202. One or more spots or stains 3/
3. Any stitching in the inflatable section of the life preserver.	203. Any cut edge of the uncoated nylon fibrous materials not seared or containing sharp edges.
4. Inflation assembly or manifold stem assembly not in accordance with requirements; not located as specified, or inoperable.	204. Any snap fastener missing, mislocated, improperly clinched, or otherwise defective. 3/ 4/
5. Any inflatable section seam construction which does not meet the minimum requirements specified.	205. Oral inflation assembly or components not specified type; oral valve not locked in the closed position.
6. Any seam tape not applied to seams, as specified. 2/	206. Any actuating lever not safety tied to the inflator assembly, as specified.
7. Any channels or voids in any cementing operation. 1/	207. Any metal component improperly finished or containing nicks, burrs, dents, sharp edges or rough surfaces. 3/
8. Inflatable section and belt assembly not joined, as specified.	208. Color of any component, not as specified.
9. Oral inflation assembly mislocated, bent, distorted, or inoperable. Oral tube clamp not installed securely. 2/	209. Any required markings missing, illegible, incomplete, incorrect or improperly located.
10. Any component, component part, or required operation omitted; or any operation improperly performed, not herein classified. 2/	210. Any inspection record patch or warning tag missing, illegible, incomplete, incorrect, or improperly attached or located.
11. Any component not as specifed, or any defect of a component or assembly not herein classified. 2/	211. Excessive wrinkling in any cementing operation.
12. Belt assembly damaged, bruised, abraded, containing imperfections, or not in accordance with requirement. 2/	212. Any inflatable section seam construction which exceeds the maximum requirements specified.

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TABLE IV. Classification of defects for visual examination of the life preserver (see 4.8.2.1). - Continued

Critical	Minor
13. Any seam not butted, as specified. 14. Any butted seam, seam tape applied to butted seams, or lapped seams of the inflatable section separating.	213. Cement on the cloth surfaces around the patches, attachments, seams, seam tapes, not visible or in excess of requirements. 3/ 214. Any clot or mass of adhesive. 215. Overlap of any seam tape in excess of requirements. 216. Any seam tape, patch, or attachment separating. 3/ 217. Belt assembly and pouch not joined, as specified. 218. Life preserver not properly inserted in the pouch.

1/ The defect shall be classified critical if it is in the inflatable section; otherwise, it is to be classified minor.

2/ The defect shall be classified critical when it seriously affects serviceability or function; otherwise, it is to be classified minor.

3/ The defect shall be classified as minor when it does not seriously affect serviceability or function; otherwise, it is to be classified critical.

4/ The snap fasteners shall be checked for proper function and attachment.

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TABLE V. List of defects for dimensions of the assembled preserver (see 4.8.2.2).

Examine	Defect
Measure the life preserver, pouch, storage container and all visible components.	Any measurement deviating from 4.8.2.2 shall be enumerated as a dimensional defect.

TABLE VI. List of defects for packaging, packing and marking (see 4.8.2.3).

Item	Defect
Exterior and interior markings	Missing, incorrect, incomplete, illegible, of improper size, location, sequence or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials	Any non-conforming component; any component missing, damaged or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, container flaps, loose strappings, etc.; bulging or distortion of the containers; unit container contains metal fastenings or stitches.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements; item not folded as specified.

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INCH–POUND

NOTICE
OF VALIDATION

MIL–L–81787B(AS)
NOTICE 1
29 October 1991

MILITARY SPECIFICATION

LIFE PRESERVER, INFLATABLE, UTILITY, TYPE LPP–1A

MIL–L–81787B(AS), dated 9 June 1983, has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Navy – AS

AMSC N/A FSC 4220

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